

SARITA

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Education

IIT Bombay, India <i>Master of Technology in Geoinformatics</i>	Jul 2024 – Present CGPA: 9.21/10
IIT Delhi, India <i>Master of Science in Physics</i>	Aug 2022 – May 2024 CGPA: 7.98/10
RPS Degree College, India <i>Bachelor of Science in Physics</i>	Jul 2019 – Jun 2022 Percentage: 86.81%

Skills

Programming & Algorithms: Python, C/C++, R, SQL, Data Structures & Algorithms
Data Science & Data Analysis: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, ANN, CNN, MySQL, Matplotlib, Seaborn, Plotly
Technologies & Tools: Git, GitHub, GIS, ArcGIS, QGIS, Rasterio, Shapely, Envi, Snap, Geomatica

Experience

Indian Institute of Technology Delhi, India <i>M.Sc. Project Student</i> 	Aug 2023 – May 2024
Designed phase retrieval system achieving 3% PSF error using gradient descent/HIO algorithms, simulating telescope optics via hexagonal mirror analysis in Python.	

Projects

Context combined vision transformers for satellite image analysis	Feb 2025 - Present
Improved vehicle detection accuracy by 10.1% in high-res satellite images by integrating contextual transformers (CoTC3) and space-to-depth techniques into YOLOv5, achieving up to 6.3% gain over baseline models.	
Super-Resolution of Gaming Images: Custom SRCNN Implementation 	April 2025
Developed and optimized a Super-Resolution CNN (SRCNN) in PyTorch to reconstruct high-resolution gaming images from low-resolution inputs. Integrated mixed-precision training, TorchScript model compilation, and efficient data pipelines for enhanced performance.	
Deep Learning Based Scene Recognition 	Jan 2025 - Present
Implemented scene recognition using Bag of Words with SIFT and k-means, analyzing accuracy across codebook sizes. Trained a 3-layer MLP on BoW features and downsampled images with various hidden sizes and activations. Built a CNN from scratch with CAM for interpretability	
Machine learning model for Ebola Outbreak Prediction 	Jan 2025
Predicted COVID-19 spread using XGBoost and Kriging (RMSE: 33.48) by incorporating population density, temperature, and health data to enhance accuracy and guide outbreak response planning.	
Flood Relief App Development 	Oct 2024 - Nov 2024
Developed flood relief app using geospatial mapping and real-time location data to optimize emergency resource allocation during disasters.	

Positions of Responsibility

Interview Coordinator, IIT Bombay	Oct 2024 -Present
Culinary Art Coordinator, IIT Bombay	Sep 2024 - Present
Secretary, Student Affairs Council, Nalanda Hostel, IIT Delhi	May 2023 - May 2024

Achievements & Awards

Academic Scholarship: National Scholarship under central sector scheme	2019 - 2024
CSIR-NET Physical Science: All India Rank - 84 in 35,721 candidates	2024
GATE Physics: All India Rank - 771 in 20,258 candidates	2024
IIT-JAM Physics: All India Rank - 124 in 12,470 candidates	2022